

Department of Computer Science
Gujarat University
MSc AI&ML – Semester I
Mathematical Foundation – Assignment (I)
Unit-1 (Part-A)
Introduction to Set theory

Submission Deadline: 15th September 2026

Question 1. Which of the following statements are true or false? Justify your answer.

- 1) The set $A = \{a\}$ and $b = \{\{a\}\}$ are equal.
- 2) $\{a\} \subset \{a\}$
- 3) $\{a\} \in \{a\}$
- 4) $\emptyset \subset \{a\}$
- 5) $\emptyset \in \{\emptyset\}$
- 6) $\{3, 4\} \subset \{1, 2, 3, \{3,4\}, 5\}$

Question 2. Write down the following sets into roaster form:

- 1) $A = \{x \in \mathbb{R} \mid x \text{ is a real number such that } x^2 = 4\}$
- 2) $B = \{x \in \mathbb{N} \mid x \text{ is a prime number less than } 20\}$
- 3) $C = \{x \in \mathbb{Z} \mid x \text{ is a positive and whose square is less than } 50\}$
- 4) $D = \{x \in \mathbb{Z} \mid x \text{ is a positive divisor of } 20\}$
- 5) $E = \{x \in \mathbb{Z} \mid x \text{ is a multiple of } 4\}$

Question 3. State which of the following sets are finite and which are infinite:

- 1) $A = \{x \in \mathbb{N} \mid x \text{ is a prime number}\}$

2) $B = \{x \in \mathbb{N} \mid x \text{ is an even}\}$

3) $C = \{x \in \mathbb{N} \mid (x + a)(x - b) = 0\}$, where a and b are arbitrary real constant

4) $D = \{x \in \mathbb{Z} \mid x^2 = 9\}$

5) $E = \{x \in \mathbb{B} \mid 3x - 1 = 0\}$

Question 4. Write down any four properties of set operations. Explain each of its through different examples.

Question 5. State and prove De Morgan's laws. Also, sketch the proof with the help of Venn diagram.

Question 6. (Application of a power set in bit representation)

Consider the set $S = \{a_1, a_2, a_3, \dots, a_{10}\}$. Each subset of given set S is denoted by the symbol Bi , where ' i ' represents a sequence of bit in binary digits. Determine the sequence of bit and corresponding set for the set $B111$ and $B73$. How will you designate the set $\{a_2, a_6, a_9\}$ and $\{a_1, a_9\}$?

Question 7. Define Cartesian product of two set.

Consider $A = \{1, 2, 3\}$, $B = \{3, 5\}$ and $C = \{2, 3, 5\}$. Then compute the following:

1) $A \times B, B \times A$

2) $A \times A, B \times B$

3) $A \times (B \cap C), A \times (B \cup C)$

4) $(A \times B) \cap (A \times C), (A \times B) \cup (A \times C)$

Question 8. (Application in Counting number of elements)

In survey of 1000 persons for usage of social media platform, it was found that 280 uses Facebook (F), 300 uses twitter (T) and 420 uses instagram (I), 80 uses both F and T, 100 uses F and I, 50 uses T and I and 30 uses all three platforms. Answer the following:

- 1) How many use at least one of these platforms?
- 2) How many use none of three platforms?
- 3) How many use only Facebook?
- 4) How many use Facebook and Twitter but not Instagram

Question 9. In a survey of 1000 individuals: 400 use Google Meet (G) 350 use Zoom (Z) 300 use Microsoft Teams (M) 150 use both G and Z 120 use both Z and M 100 use both G and M 60 use all three Answer the following:

- 1) How many use exactly two platforms?
- 2) How many use Google Meet or Zoom but not Teams?
- 3) How many uses at least one platform but not all three?

Question 10. In a college tech club of 800 students: 300 enrolled in AI course (A) 350 enrolled in Web Dev course (W) 250 enrolled in Cybersecurity course (C) 100 students are in both AI and Web Dev 80 students are in both AI and Cybersecurity 90 students are in both Web Dev and Cybersecurity 50 students are in all three. Answer the following:

- 1) How many students are enrolled in exactly one course?
- 2) How many students are enrolled in at least two courses?
- 3) How many students are not enrolled in any course?

Question 11. If $A = \{x \in \mathbb{N} / x < 6\}$ and $B = \{x \in \mathbb{N} / (x+1)^2 < 49\}$ then prove that $A = B$

Question 12. If $A = \{a, b, c, d, e\}$ and $B = \{d, e, f, g\}$, then obtain $n(A)$, $n(B)$, $n(A \cap B)$ and $n(A \cup B)$.

Question 13. If $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 5, 6, 8\}$ and $C = \{1, 6, 7\}$, then prove that

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C).$$

Question 14. The results of students of two classes of a school with 120 students in the subjects of Mathematics and Physics are as follows:

- (i) The number of students passing in Mathematics is 55.
- (ii) The number of students passing in Physics is 60.
- (iii) The number of students passing in both the subjects is 22.

Answer the following:

1. How many students are passing only in Mathematics?
2. How many students are passing only in Physics?
3. How many students are failing in both the subjects?

Question 15. There are 63 students in a class of a school. Out of these, 15 students do not play hockey or football. 35 students play football and 20 students play hockey.

Answer the following:

1. How many students play both the Games?
2. How many students play only football ?
3. How many students play only hockey ?

Part-B

Coordinate Geometry

Question 0. Plot the points A(2, 3), B(-4, 1), C(0, -5) and D(3, -2) on the Cartesian plane. State which quadrant each point lies in.

Question 1. Derive the Distance formula for two points in the Cartesian plane.

Question 2. Prove that the points A(4, 4), B(3, 5) and C(-1, -1) are the vertices of a right-angled triangle.

Question 3. If the distance between the points A(5, x) and B(2, 6) is $3\sqrt{2}$, then find the value of x.

Question 4. Show that the vertices of the triangle (7, 9), (3, -7) and (-3, 3) form a right-angled isosceles triangle.

Question 5. Show that the points (1, 0), (5, 3), (2, 7) and (-2, 4) are the vertices of a rhombus.

Question 6. Show that (-1, 0), (2, 3), (4, 1) and (1, -2) are the vertices of a rectangle.

Question 7. Find the point on the Y-axis which is equidistant from the points (-5, -2) and (3, 2).

Question 8. Find the area of the triangle whose vertices are (4, 4), (3, -2) and (-3, 16).

Question 9. For which value of x is the area of the triangle formed by the vertices (x, 4), (8, 2) and (6, 7) equal to 13 units?

Question 10. Find the coordinates of the point which divides the line segment joining the points (6, 3) and (-4, 5) in the ratio 3:2 (i) internally and (ii) externally.

Question 11. Find the ratio in which P(-1, -1) divides line segment AB, where A(4, 4), B(7, 7).

Question 12. For A(-2, 3) and B(3, 0), find the ratio in which the Y-axis divides AB from A's side.

Question 13. Show that the points (1, 1), (2, 3) and (3, 5) are collinear.

Question 14. Without using the Pythagoras theorem, show that the points A(4, 4), B(3, 5) and C(-1, -1) are the vertices of a right-angled triangle.

Question 15. If the angle between two lines is $\pi/4$ and the slope of one of the lines is $1/2$, find the slope of the other line.

Question 16. Obtain the measure of the angle between the following pairs of lines:

1. $x - y + 3 = 0, y - 2 = 0$
2. $x - y + 4 = 0, 5x - y + 3 = 0$
3. $x + y + 1 = 0, x - y = 0$

Question 17. Find the equation of the perpendicular bisector of the line segment joining the points A(2, 3) and B(6, -5).

Question 18. Find the equation of the line passing through the points (2, 3) and (5, -2).

Question 19. Find the equation of the line which passes through the point (-5, 4) such that the portion intercepted between the axes is divided by the point in the ratio 1 : 2.

Question 20. Find the equation of a line through the intersection of $x - y - 1 = 0$ and $2x - 3y + 1 = 0$, such that it:

1. Has slope -2
2. Is parallel to the line $x + y + 4 = 0$
3. Passes through (1, 2)

Question 21. Convert the equation $3x - 4y + 12 = 0$ into slope-intercept form and intercept form. Hence find its slope, y-intercept, and x-intercept.

Question 22. Find the midpoint of the line segment joining the points A(6, -3) and B(-2, 7). Also verify that this midpoint lies on the line joining A and B.

Question 23. Find the perpendicular distance of the point (3, -5) from the line $3x + 4y - 26 = 0$.

Question 24. Find the equation of the line passing through the point (2, 3) and perpendicular (normal) to the line $4x - 3y + 7 = 0$.

Question 25. The following data shows values of x and y :

x : 1, 2, 3, 4, 5

y : 2, 4, 5, 4, 5

Using the Least Square Method, find the slope (m) and intercept (c) of the best-fit line $y = mx + c$. Hence write the equation of the regression line.

Question 26. Find the equation of the plane passing through the point (1, 2, 3) and having direction ratios of its normal as (2, -1, 4).

Question 27. Find the radius of the circle whose centre is at (0, 0) and which passes through the point (-6, 8).

Question 28. The co-ordinates of the mid-point of a segment are (2, 3). If co-ordinates of one of the end points of the line segment are (6, 5), find the co-ordinates of the other end point

Question 29. The co-ordinates of the vertices of a triangle are (3, -1), (10, 7) and (5, 3). Find the co-ordinates of its centroid.

Question 30. Determine the distance from point $P(-6, 4)$ to the line with equation $5x - 3y + 15 = 0$.

Question 31. The following data shows the marks obtained by students in a class test (x) and their corresponding scores in the final exam (y):

x : 2, 4, 6, 8, 10

y : 3, 7, 5, 10, 12

Using the Least Square Method, find the slope (m) and intercept (c) of the best-fit line $y = mx + c$.